

Week 2 – Day 2: Define Success Criteria, User Stories, and Expected Lab Use Cases

Task: Define success criteria, user stories, and expected lab use cases for the cybersecurity home lab.

Elaboration:

Establish measurable **success criteria** that determine whether the home lab meets its intended purpose. The lab should successfully allow the user to deploy and connect the virtual machines through **Oracle VirtualBox**, configure **pfSense as the firewall**, establish an isolated **DMZ**, and use **Kali Linux** to perform controlled security testing against **Metasploitable2**. The lab should also allow **Wireshark** to capture and analyze network traffic generated during these activities.

Evaluation Criteria:

The project will be considered successful if:

1. The virtual lab is fully deployed and all virtual machines communicate as intended.
2. Wireshark successfully captures and displays network traffic for analysis.
3. Nmap identifies active hosts and services within the lab environment.
4. Controlled Metasploit demonstrations execute successfully in the isolated environment.
5. pfSense blocks unauthorized traffic according to the configured firewall rules.
6. Network segmentation separates the internal LAN and DMZ effectively.
7. All project documentation, diagrams, screenshots, and reports are complete and organized.
8. The project can be demonstrated from start to finish, showing planning, implementation, testing, and security improvements.

Develop **user stories** that describe how the intended user will interact with the lab. For example, the user should be able to conduct network reconnaissance, identify vulnerabilities, perform controlled penetration-testing activities, capture network traffic, analyze security events, and document findings.

Define the primary **use cases** the lab is expected to support, including:

- Network discovery and reconnaissance
- Vulnerability scanning and assessment
- Controlled penetration testing
- Firewall and DMZ security testing
- Packet capture and network traffic analysis
- Identification of suspicious or malicious network activity
- Documentation and reporting of security findings